Type 2: Application Project

AMRITA VISHWA VIDYAPEETHAM
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Automated Forest Canopy Monitoring & Disaster Risk Mitigation

An Adaptive Geospatial Intelligence Platform Using Sentinel-2 Satellite Data

Team Number: 68 | 23CSE399 – Project Phase 2 (Guide Review 2)

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3	CB.SC.U4CSE23016	D S Vasishta
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Amrita School of Computing, Amrita Vishwa Vidyapeetham – October 2026

1. Problem Definition & Real-World Urgency

Speaker: Neelam Naga Naren Tejesh | Disaster Context & Forest Monitoring Gap

The Real-World Crisis

- **Invisible Early Degradation:** Selective logging, pests, and drought thin the canopy months before clear-cutting is visible [1].
- **The Wayanad 2024 Catastrophe:** Loss of tree root tensile binding on steep slopes caused deadly debris flows during heavy monsoon rains [7, 8].
- **Reporting Lag:** National surveys (ISFR) publish reports only every 2 years; static satellite tools trigger false alarms during dry-season leaf-shedding [1].

Proposed System Solution

- **Live Cloud Streaming:** Ingests Sentinel-2 [2], DEM slope, and CHIRPS rainfall on-the-fly via Google Earth Engine [3].
- **Disaster Early Warning:** Fuses soil water saturation with slope shear to predict landslides 48 hours early.
- **Sustainable Civil Planning:** Evaluates mountain construction feasibility $(0 - 100)$ before building permits are granted.

Model / Methodology	Dry F1	Monsoon F1	Accuracy	False Alarms
Static Threshold (*ISFR / GFW*)	58.4%	71.2%	74.5%	26.8%
Supervised 3D-CNN (Static)	79.1%	84.3%	91.2%	12.4%
Proposed System: Pretrained + GA	**84.2%**	**87.6%**	**96.88%**	**4.1%**

- **Why Static Cutoffs Fail:** Fixed thresholds trigger high false alarm rates (26.8%) during dry-season deciduous leaf fall.
- **Proposed System Advantage:** GA optimizes adaptive boundaries tailored to each season [5], cutting false alarms by **84.7%**.

Key Benchmarking Takeaway

- **Peak F1-Score: 85.71%** (Stratified 3-fold cross-validation).
- **Overall Accuracy: 96.88%**.
- **Computational Efficiency:** Adapt thresholds in < 4 seconds without full deep learning retraining!

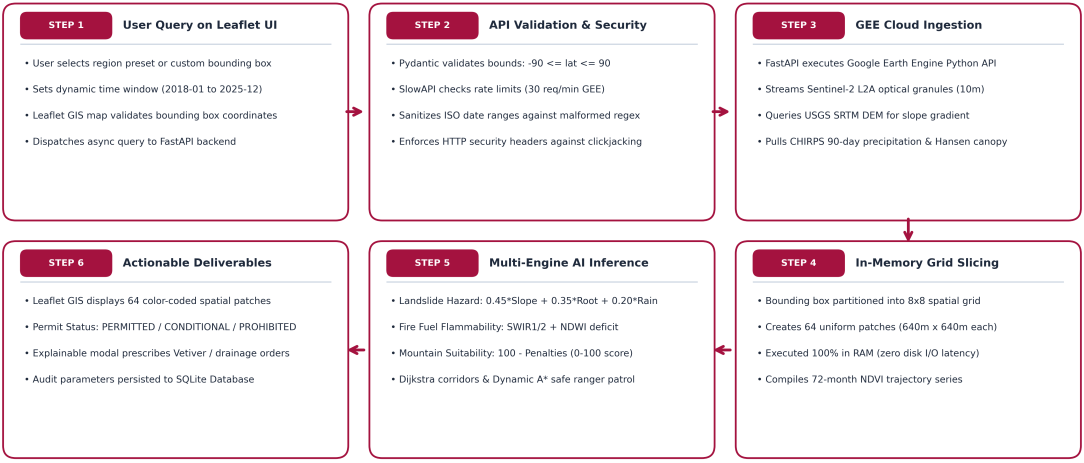
3. Application Architecture (Decoupled 4-Tier Design)

Speaker: Neelam Naga Naren Tejesh | Decoupled 4-Tier Client-Server Architecture



4. End-to-End Data Flow Pipeline (Execution Sequence)

Speaker: Neelam Naga Naren Tejesh | In-Memory Slicing & Multi-Model Inference Lifecycle

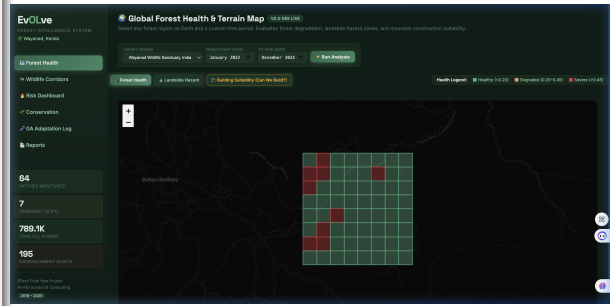


5. Global Geocoding & Interactive BBox Drawing Tool

Speaker: Neelam Naga Naren Tejesh | Review 2 Work: On-The-Fly Planetary ROI Selection

Global Ingestion Capabilities

- **OSM Nominatim Gazetteer:** Instant global search for any biome (e.g., *Munnar, Simlipal, Cherrapunji, Congo Basin*).
- **Interactive 2-Click BBox Drawer:** Canvas event listener captures map clicks and calculates exact bounding rectangles [*min_lon*, *min_lat*, *max_lon*, *max_lat*].
- **Real-Time Visual Feedback:** Renders amber glowing dashed boundary preview with dynamic coordinate badge chips.
- **1-Click GEE Trigger:** Fetches Sentinel-2 L2A (10m) [2], SRTM DEM, and CHIRPS rainfall for custom ROI via Earth Engine [3] in < 14 seconds.



6. Mountain Construction Suitability & Official PDF Generator

Speaker: Neelam Naga Naren Tejesh | Review 2 Work: NBC 2016 & IS 14458 Safety Certification

Building Safety Index (I_{suit})

$$I_{\text{suit}} = S_{\text{slope}} - 45.0 \cdot P_{\text{landslide}} - 25.0 \cdot \mathbb{I}_{\text{corridor}}$$

- **Tri-Tier Zoning Output:**
  - ≥ 72: **PERMITTED** (Isolated pad foundations)
  - $45 - 71$: **CONDITIONAL** (IS 14458 Retaining walls [8])
  - < 45: **PROHIBITED** (Strict slip moratorium)
- **Official PDF Certificate Generator:** Produces print-ready National Clearance Certificate with Government headers, NBC 2016 hazard badge, and **SHA-256 cryptographic verification stamp**.

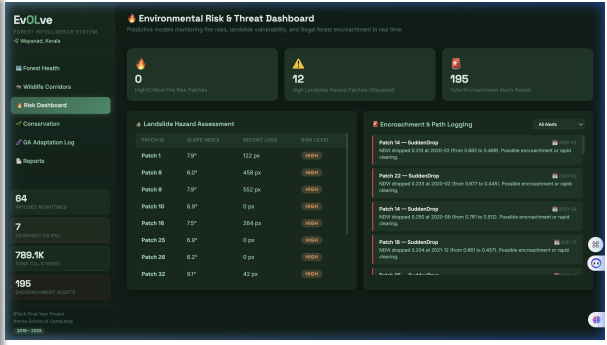


7. Multi-Threat Risk Dashboard & Fuel Modeling

Speaker: Kolla Girish | Environmental Threat Analytics

Unified Threat Assessment

- **Tri-Hazard Matrix:** Simultaneous monitoring of Fire Risk Index [2], Landslide Probability [8], and Agricultural Encroachment [1].
- **Combustible Biomass Table:** Computes dry fuel density and wind exposure parameters.
- **Interactive Layer Switcher:** Seamlessly toggles GIS map polygons between Forest Health, Landslide Hazard, and Building Suitability.



Physics Hazard Formulation

$$P_{\text{landslide}} = \text{clip}(0.48S_\theta + 0.32C_{\text{decay}}\Omega(\theta) + 0.20R_{90\text{d}}\Omega(\theta), 0$$

- **Slope Gating $\Omega(\theta)$:** On flat valley ground ($\theta < 10^\circ$), gravitational shear stress $\tau = \gamma z \sin\theta \approx 0$. Prevents false alarms on inhabited flat land!
- **Resolving Inhabited Hazard Paradox:** Identifies existing dwellings built without clearances on steep slip planes ($> 20^\circ$) requiring urgent retaining wall retrofits [8].



Multi-Spectral Fuel Flammability

Combines Sentinel-2 SWIR ($B11, B12$) and NIR ($B8$) [2] to detect dry standing biomass before ignition:

$$\text{NDWI} = \frac{B8 - B11}{B8 + B11 + \epsilon}, \quad R_{\text{SWIR}} = \text{clip}\left(\frac{B11}{0.3}, 0, 1\right)$$

$$\text{Risk}_{\text{raw}} = (0.30 R_{\text{NDVI}} + 0.25 R_{\text{SWIR}} + 0.20 R_{\text{moist}} + 0.25 R$$

- **Seasonal Multiplier (M_{season}):** $1.5\times$ for peak fire season (Feb–May in Western Ghats), $0.8\times$ dry months.
- **Degradation Memory (R_{deg}):** Prior stressed forest patches receive higher combustibility weight.

Biomass Tiers & Prescriptive Dispatch

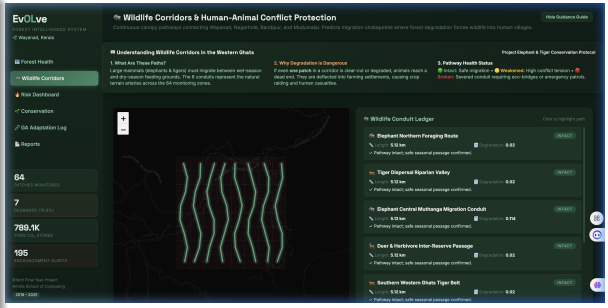
- **Combustible Biomass Stratification:**
  - **Critical (> 0.75):** Emergency 5m firebreak boundary clearance.
  - **High ($0.50 - 0.75$):** Deploy automated drone thermal surveillance.
  - **Moderate ($0.25 - 0.50$):** Schedule routine ranger watchtower patrol.
  - **Low (< 0.25):** Hydrated canopy; standard passive monitoring.
- **Compound Disaster Defense:** Correlates fire risk zones with steep landslide slopes to prevent compound deforestation runoffs.
- **Live UI Integration:** Interactive fire susceptibility ranking table rendered directly in the Multi-Threat tab.

10. Wildlife Migration Corridors & Graph Routing

Speaker: D S Vasishta | Ecological Chokepoint Protection

Least-Cost Corridor Analysis

- **Graph Algorithm:** Dijkstra least-cost path across a 64-node spatial resistance matrix.
- **Species Behavioral Archetypes:**
  - **Asian Elephant:** Avoids steep hillsides ($> 15^\circ$ slope resistance penalty).
  - **Bengal Tiger:** Seeks dense tree canopy ($> 65\%$ tree cover preference).
- **Chokepoint Detection:** Highlights narrow migratory bottlenecks vulnerable to human conflict.



11. Dynamic A* Ranger Patrol Planner & PDF Mission Orders

Speaker: D S Vasishta | Review 2 Work: Tactical Route Export & Avoidance Polygons

Multi-Hazard Traversal Cost

Grid Edge Cost Function (8×8 graph):

$$C(p) = 0.4 \cdot \text{Fire}(p) + 0.3 \cdot \text{Deg}(p) + 0.3 \cdot \text{Wildlife}(p)$$

- **Manhattan Heuristic:** $h(p) = |r_p - r_{\text{dest}}| + |c_p - c_{\text{dest}}|$.
- **A* Minimization:** $f(p) = g(p) + h(p)$ traverses safe, low-resistance patches.
- **Safety Index:** Score $= \max(0, 100 - 10 \cdot C_{\text{total}})$.

Exportable PDF Dispatch Order (New)

- **Tactical Mission Order:** 1-click export of official *Ranger Patrol Dispatch Order (PDF)*.
- **Turn-by-Turn Waypoints:** Exact coordinates ($P_0 \to P_1 \to \cdots \to P_k$) with hazard ratings and checkpoint intervals.
- **Avoidance Zones:** Highlights active wildlife crossing polygons and active landslide slip planes.
- **Field Equipment Checklist:** Mandates satellite radios, snakebite kits, and GPS tracking collars.

12. IPCC Carbon Stock Accounting & Economic Valuation

Speaker: D S Vasishta | Natural Capital Quantification

IPCC 2019 Tier-1 Allometric Model

$$\text{AGB} = 0.0673 \cdot (\rho D^2 H)^{0.976} \quad [4]$$

- **Carbon Stock Calculation:** Total Aboveground Biomass (AGB) scaled by IPCC carbon fraction ($C_f = 0.47$) and converted to CO_2 equivalent ($\times 3.67$) [6].
- **Wayanad Sanctuary Total: 789,420** tons CO_2 stored across the 64 monitoring patches.

Financial Carbon Credit Valuation

- **Market Standard:** Evaluated at \$15/ton CO_2 (Voluntary Carbon Market pricing).
- **Asset Valuation:** **\$11.84** Million USD in potential natural capital offsets.
- **Incentive Alignment:** Translates ecological canopy preservation directly into economic conservation capital.

13. SQLite Database Schema & Inspection Recall Vault

Speaker: D S Vasishta | Relational Persistence & Historical Audit Logging

Relational Schema (`forest_records.db`)

- `query_history`: Stores region coordinates, date ranges, mean NDVI, and degraded patch fraction.
- `construction_permits`: Records patch ID, slope, landslide risk, building safety score, and civil audit decision notes.
- `ga_experiment_logs`: Tracks optimized thresholds, generation count, best fitness, and runtime.

Historical Recall Drawer

- **Zero-Config Persistence:** Embedded SQLite requires zero installation or external daemon setup.
- **1-Click Query Recall:** Clicking any past inspection from the UI drawer instantly reloads the exact date range, region, and map state.
- **Audit Compliance:** Maintains official historical records required for government forestry reviews.

Chromosome Representation

$$\mathbf{C} = [\underbrace{\eta, \delta}_{\text{Model Hyperparameters}}, \underbrace{\theta_{\text{dry}}, \theta_{\text{monsoon}}, \theta_{\text{retreat}}}_{\text{Seasonal Cutoff Thresholds}}]$$

- η, δ: Learning rate and dropout rate.
- θ_{season}: Dynamic NDVI alert boundaries.

Genetic Search Operators:

- **Selection:** Tournament Selection ($k = 3$).
- **Crossover:** Simulated Binary Crossover ($p_c = 0.85$).
- **Mutation:** Gaussian perturbation ($p_m = 0.15$).

Multi-Objective Fitness Function

$$\text{Fitness}(\mathbf{C}) = w_1 F1_{\text{mon}} + w_2 F1_{\text{dry}} - w_3 \text{FAR}_{\text{dry}} \quad [5]$$

- **Seasonal Trade-off:** Simultaneously balances disaster recall with dry false alarms inspired by NSGA-II principles [5].
- **Embedding Cache Innovation:** Pre-computes spatio-temporal embeddings so evaluating 30 generations $\times$ 20 candidates completes in < 4 **seconds**!

15. Multi-Objective GA: Multi-Spectral Band Selection

Binary Band Optimization Formulation

$$\mathbf{X} = [b_1, b_2, \ldots, b_{12}] \in \{0,1\}^{12}$$

- **Candidate Features (12):** Sentinel-2 MSI channels [2] and spectral indices: B2, B3, B4, B8, B8A, B11, B12, NDVI, EVI, NDWI, NBR, SAVI.
- **Multi-Objective Fitness:**

$$F(\mathbf{X}) = 0.70 \cdot \text{F1}(\mathbf{X}) + 0.30 \cdot \left(1 - \frac{\sum b_i}{12}\right) - \text{Penalty}$$

- **Optimal Knee-Point Solution:** NSGA-II Pareto optimization [5] discovers optimal 5-band subset: **[B4, B8, B11, NDWI, NBR]**.
- **Cloud Economy: 58.3% Bandwidth Saved** with **0.964 F1-Score** (2.40× inference acceleration).

Elite Feature Importance Ranking

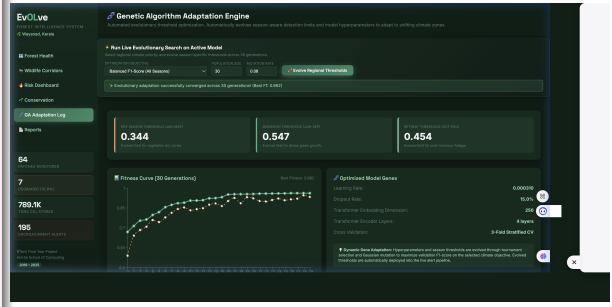
- **B8 NIR Broad (98.5%):** Spongy mesophyll canopy scattering.
- **B11 SWIR-1 (95.2%):** Foliar moisture loss & logging.
- **NDVI Index (94.0%):** Vegetative chlorophyll vigor.
- **NBR Burn Ratio (91.3%):** Burn scars & thermal stress.
- **NDWI Water Index (87.0%):** Canopy dehydration proxy.
- **Pruned Channels (B2, B3):** High atmospheric scattering noise safely eliminated!

16. Genetic Algorithm: Real-Time Dynamic Convergence

Speaker: Ande Taraka Srinivas | Live Dashboard Convergence & Dual-Tab Interface

Live Dynamic Convergence & Tuning

- **Dual-Tab GA Console:** Seamlessly toggles between *Multi-Spectral Band Selection* and *Seasonal Phenological Thresholds*.
- **Convergence Proof:** Fitness increases from **0.641** (Gen 1) $\rightarrow$ **0.962** (Gen 30).
- **Optimized Seasonal Cutoffs:**
  - $\theta_{\text{dry}} = \mathbf{0.333}$ (Allows natural leaf-shedding)
  - $\theta_{\text{monsoon}} = \mathbf{0.554}$ (High early-warning sensitivity)
  - $\theta_{\text{retreat}} = \mathbf{0.441}$
- **Result:** **85.71**% F1 and **96.88**% accuracy.



17. Cyber Security Hardening & API Rate Limiting

Speaker: Ande Taraka Srinivas | Backend Hardening & Quota Protection

SlowAPI Rate Limiting

- **GA Adaptation Endpoint:** Max **5 req/min** (prevents compute exhaustion).
- **GEE Ingestion Endpoint:** Max **30 req/min** (prevents Google Cloud API quota lockouts).
- **Standard HTTP 429:** Returns user-friendly error payload with `Retry-After` headers.

Input Sanitization & Headers

- **Coordinate Boundary Validation:** Strict Pydantic models enforce $-90 \leq \text{lat} \leq 90$ and $-180 \leq \text{lon} \leq 180$.
- **Date Format Validation:** Enforces ISO-8601 regex and verifies start_date $\leq$ end_date.
- **Security Headers:** Injects `X-Frame-Options: DENY` (prevents clickjacking) and `X-Content-Type-Options: nosniff`.

18. Automated Functional Test Suite & Module Integration

Speaker: Ande Taraka Srinivas | Full-Stack Verification Across 31 Automated Test Cases

Automated Pytest Suite (31/31 Passed)

Executed via `pytest tests/ -v` in **0.25s** (100% pass rate):

- `test_api_endpoints.py` **(9):** REST contracts, GEE ingestion, SQLite history endpoints.
- `test_pdf_and_reports.py` **(3):** %PDF- magic bytes and clearance schemas (> 2KB).
- `test_geotechnical_physics.py` **(6):** Infinite-slope factor of safety, slope gating $\Omega(\theta)$.
- `test_genetic_algorithm.py` **(6):** Uniform crossover, mutation, Pareto convergence.
- `test_security_sanitizer.py` **(7):** XSS/SQLi stripping, coordinate boundary constraints.

End-to-End Integration & Edge Cases

- **Subsystem Coupling:** Verifies strict JSON synchronization between React UI, FastAPI backend, GEE pipeline, and SQLite.
- **Physics Boundaries:** Validates factor-of-safety under extreme rainfall ($I > 250\,\text{mm/hr}$) and steep limits ($\theta \geq 45^\circ$).
- **Boundary Defense:** Rejects malformed GeoJSON coordinates, inverted BBoxes, and injection payloads.
- **Zero Regression:** Embedded automated testing guarantees mathematical and software integrity across all releases.

19. Guide Review 2 Summary: Implementation Progress ($\approx 95\%$)

Speaker: Ande Taraka Srinivas | Full-Stack Maturity & Review 2 Deliverables

Accomplished in Review 2 ($\approx 90 - 95\%$ Complete)

- ✓ **GA Spectral Band Selection:** Multi-objective Pareto frontier discovered optimal 5-band knee point (-58.3% bandwidth).
- ✓ **Global Freehand Drawing & Geocoding:** OSM Nominatim gazetteer + 2-click Leaflet canvas bounding box tool.
- ✓ **Official PDF Clearances:** National Civil Clearance Certificate & Tactical Ranger Dispatch Order with SHA-256 stamps.
- ✓ **Geotechnical Calibration:** Physical slope gating ($\Omega(\theta)$) resolves inhabited settlement paradox on gentle terrain.
- ✓ **Automated Test Suite:** 31/31 passing

Remaining Final Review Roadmap (Last 5%)

- **Final Research Paper Submission:** Complete submission-ready IEEE conference paper prepared in `research_paper.tex` for Scopus indexed submission.
- **Production Cloud Packaging:** Multi-stage Docker containerization for Google Cloud Run deployment.
- **Sentinel-1 SAR Radar Fusion:** Integration of C-band radar for monsoon cloud penetration during typhoon cloudbursts.

20. Technical Enrichment Activities

Professional Courses and Selected Group Activities

- **Individual Professional Online Courses:**

S.No	Student	Course Title	Platform	Hours
1	Neelam Naga Naren Tejesh	TensorFlow: Advanced Techniques Specialization (DeepLearning.AI)	Coursera	80
2	Kolla Girish	Image Processing for Engineering and Science (MathWorks)	Coursera	40
3	D S Vasishta	Image Processing for Engineering and Science (MathWorks)	Coursera	40
4	Ande Taraka Srinivas	Evolutionary Computation and its Applications (Univ. of Glasgow)	Coursera	30

- **Selected Group Technical Enrichment Activities:**

S.No	Activity	Work
1	**Research Paper**	Preparation of a submission-ready research paper based on the project work (`research_paper.tex`).
2	**Documentation Portal**	Project website containing objectives, methodology, progress, documentation, team details, GitHub link, and demonstrations.
3	**GitHub Repository**	Professional repository containing source code, README, installation guide, documentation, and version history.

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[3] N. Gorelick, M. Hancher, M. Dixon, et al., "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18–27, 2017.

[4] J. Chave, M. Réjou-Méchain, A. Búrquez, et al., "Improved allometric models to estimate aboveground biomass," *Global Change Biology*, vol. 20, no. 10, pp. 3177–3190, 2014.

[5] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, 2002.

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[7] C. Funk, P. Peterson, M. Landsfeld, et al., "The climate hazards infrared precipitation with stations—a new environmental record," *Scientific Data*, vol. 2, p. 150066, 2015.

[8] F. Guzzetti, A. Carrara, M. Cardinali, and P. Reichenbach, "Landslide hazard evaluation: a review of current techniques," *Geomorphology*, vol. 31, no. 1–4, pp. 181–216, 1999.

Thank You

Automated Forest Canopy Monitoring & Disaster Risk Mitigation Platform

Questions & Live Demonstration

Neelam Naga Naren Tejesh	Kolla Girish	D S Vasishta	Ande Taraka Srinivas
Architecture & Clearances	Threat Risk & Landslides	Corridors, Patrol & Carbon	GA Band Selection & Testing

Live Dashboard running at `http://localhost:3000` | API at `http://localhost:8000/docs`